



Sectoral Foreign Direct Investment Inflows and Stock Market Returns in Türkiye: Evidence From Borsa İstanbul Sector Indices



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Abstract: Whether foreign direct investment (FDI) inflows provide predictive information for stock market returns at the sectoral level remains insufficiently understood, particularly in emerging markets. This study examines the short-term predictive relationship between sectoral FDI inflows and stock market returns across major sectors of the Turkish equity market. Monthly data covering January 2010 to November 2025 were analysed for the banking, finance and insurance, services, manufacturing, industrial, and wholesale and retail sectors. Separate Vector Autoregression (VAR) models were specified for each sector, with Borsa İstanbul 100 (BIST 100) index returns and USD/TRY exchange-rate returns included as control variables to account for broad market and exchange-rate conditions. Lagged predictive relationships were assessed using Granger causality tests, while the dynamic responses of the variables to shocks were examined through impulse response analysis. Model adequacy was evaluated using standard diagnostic tests, and substantive interpretation was restricted to the banking and finance and insurance models that satisfied the required diagnostic criteria. No statistically significant Granger-predictive relationship was identified in either direction between sectoral FDI inflows and the corresponding sectoral stock market returns in either of these diagnostically adequate models. The impulse response results likewise provide limited evidence of a persistent or systematic transmission from sectoral FDI inflows to sectoral stock market returns. Overall, the findings suggest that sector-specific FDI inflows should not be regarded as a robust short-term predictor of sectoral equity returns in Türkiye over the sample period. The results also indicate that sectoral stock market dynamics may be driven more strongly by broader market conditions and other macro-financial factors than by contemporaneous changes in sector-specific FDI inflows.

Keywords: Foreign direct investment; Sectoral stock market returns; Borsa İstanbul; Vector Autoregression model; Granger causality test

1. Introduction

With globalization, international capital movements have become a significant element in the financing structure and financial market development of developing economies (Aydemir & Genç, 2015; Prasad et al., 2007). International capital movements consist of different sources such as external loans, portfolio investments, and foreign direct investment (FDI) (Karagöz, 2007). Among these, FDI stands out from other capital flows because it can provide long-term production capacity, technology transfer, institutional knowledge accumulation, and increased productivity (Borensztein et al., 1998). Due to these characteristics, the effects of FDI flows on economic growth, investments, and financial markets are extensively studied in the literature (Alfaro et al., 2004; Ayanwale, 2007; Har et al., 2008; Pegkas, 2015).

The reflection of FDI on stock markets can theoretically occur through direct and indirect channels. Foreign investments can boost production capacity and firm productivity, strengthening expectations for future cash flows and thus supporting firm valuations. However, the fact that FDI inflows are considered real investments and their effects emerge over time may limit their predictive power on short-term stock market returns. Therefore, the direction and strength of the relationship between FDI flows and stock returns can vary depending on the characteristics of the sector in which the investment is made and the time horizon examined. A significant portion

of studies examining the relationship between FDI and stock markets in Türkiye relate total FDI flows to general market indices such as the Borsa İstanbul 100 (BIST 100) (Aydın & Aksoy, 2023; Baydaş & Polat, 2018). This approach may not adequately reflect the sectoral distribution of FDI flows and the differences in financial market responses across sectors. Since sectors differ in capital intensity, dependence on external financing, competitive structure, and capacity to attract foreign investment, it cannot be assumed that the results obtained at the aggregate level are valid for all sectors. However, studies examining the short-term forecasting relationships between sectoral FDI flows and the returns of the relevant BIST sector indices in Türkiye are limited. This study addresses this gap by examining the FDI-stock market relationship at the sectoral level. The aim of this study is to examine the short-term dynamic and lagged forecasting relationships between sectoral FDI flows and BIST sector index returns in Türkiye. Monthly data for the period 2010:01–2025:11 were used in the analysis. FDI flows for the banking, finance-insurance, services, manufacturing, industrial, and wholesale-retail sectors, as well as the returns of the banking, insurance, services, industrial, and trade indices, were evaluated. The BIST 100 index return and the USD/TRY exchange rate return were included in the model to represent general market and exchange rate conditions. The main contribution of the study is that instead of retesting the relationship between total FDI flows and the general market index, it evaluates sectoral FDI flows and BIST sector index returns together. This study examines the short-term predictive power of FDI flows and the sector index responses to investment shocks through Granger causality tests and impulse-response analyses. The study's contribution lies not in a methodological innovation, but rather in the decomposition of the FDI-stock market relationship at the sectoral level and the highlighting of differences between sectors.

2. Theoretical Framework and Literature Review

The relationship between FDI and stock market returns can be explained through a two-way mechanism. FDI inflows can influence sector valuations through expectations regarding production capacity, productivity, and future cash flows (Alfaro et al., 2004; Borensztein et al., 1998). Conversely, strong sector performance and advanced financial markets can also create a positive signal for foreign investment decisions (Soumaré & Tchana Tchana, 2015).

2.1 Studies on Foreign Capital Movements in Türkiye and the Istanbul Stock Exchange

Türkiye has implemented regulations aimed at liberalizing capital movements, particularly after 1989 (Aydemir & Genç, 2015). This process has led to increased capital flows into Türkiye, and studies examining the relationship between foreign capital movements and the BIST have become widespread. The first group of studies focuses on the relationship between foreign portfolio investments and the BIST 100 index. These studies yield different results regarding the direction and strength of the relationship. Altan (2022), examining the relationship between foreign portfolio investments and the BIST 100, found a long-term relationship between stock market performance and foreign portfolio indicators during the 2010–2022 period. Alp et al. (2024), examining the causal relationship between foreign portfolio investor ratios and the BIST 100 index, found a unidirectional causal relationship from the BIST 100 index to the foreign portfolio investor ratio. The study found that foreign investors largely shape their market movements according to the performance of the BIST 100, and that local market dynamics are decisive in foreign capital inflows.

The second group of studies focuses on the relationship between foreign investor behavior and exchange rates, interest rates, inflation, and confidence indicators. Köycü & Ege (2023) found a bidirectional Granger causality between the number of foreign investors in BIST and the BIST ALL index. However, they identified a causality from USD/TL, inflation, interest rate, and consumer confidence index to the number of foreign investors. Aydın & Aksoy (2023) examined the relationship between portfolio investments and the dollar exchange rate and the BIST 100, stating that there are long-term relationships between these variables.

The third group of studies examines foreign investor preferences at the firm and sector levels. These studies reveal that foreign ownership is related to market volatility, firm performance, and financial indicators, but that the effects vary depending on the sector and firm characteristics. Yıldız (2017), based on panel data obtained from 192 firms during the 2006–2015 period, examined the relationship between foreign ownership at the firm level and market volatility and showed that foreign fund ownership reduced return volatility and contributed to market stability. Bozkurt (2015), using panel data regression, analyzed the micro determinants of foreign portfolio investments and revealed that financial indicators such as net profit margin, current ratio, and debt/equity ratio influenced foreign investor preferences. Similarly, Şenol & Gülcemal (2022), in their study, showed that variables such as return on investment and trading volume had significant effects on the foreign investor ratio in the BIST manufacturing industry. Atik & Yılmaz (2021), one of the limited number of studies that directly addressed the sectoral level, tested the relationship between BIST sector indices and monthly changes in foreign ownership. According to the study findings, despite high foreign ownership rates, changes in sector indices are not directly caused by changes in foreign ownership, and performance is more driven by BIST's internal dynamics and

macroeconomic factors. Doğukanlı & Çetenak (2008) examined the dynamic interaction between portfolio flows and market returns and found a causal relationship between equity returns and foreign portfolio investments. Ülkü & İkizlerli (2012), analyzing the relationship between net foreign capital flows and market returns, showed that foreign investors can react to past returns and are particularly sensitive to market movements during periods of instability. Arık (2025) revealed that macro-financial risk indicators such as credit default swap (CDS) premium, exchange rate, and interest rate are decisive in explaining foreign portfolio returns, and in impulse-response analyses, portfolio flows react negatively to CDS and exchange rate shocks and positively to interest rate shocks.

2.2 The Relationship Between Foreign Direct Investment, Financial Markets, and the Stock Exchange

FDI refers to investments by foreign investors aimed at establishing permanent control over production or operations (Karagöz, 2007). These investments are important for developing economies because, in addition to capital inflow, they can provide technology, management expertise, and production capacity (Aslan, 2023).

FDI can indirectly affect stock markets through production capacity, productivity, corporate governance, and firm valuations (Ayanwale, 2007; Borensztein et al., 1998). The literature emphasizes that FDI strengthens market liquidity by increasing capital inflow, supports firm valuations, and positively affects stock market performance through long-term economic growth (Ayanwale, 2007). In addition, FDI positively impacts stock markets by improving the financial performance of firms through technology transfer, improvement in corporate governance quality, and increased productivity (Borensztein et al., 1998). However, some studies show that this effect depends on the country's level of financial development and macroeconomic stability, and that in economies with weak financial structures, the impact of FDI on the stock market may remain limited (Alfaro et al., 2004).

Studies examining the relationship between FDI and the stock market in Türkiye mostly focus on general market indices such as the BIST 100. Baydaş & Polat (2018) examined the effect of FDI on the BIST 100 index. In the analysis, the BIST 100 index was used as the dependent variable and FDI as the independent variable. As a result of the study, it was determined that there was no cointegration relationship between the variables and, similarly, no causal relationship between the variables. Aslan (2023) used monthly data from the period 2012–2022 in his study. In the study, the BIST 100 index was used as the dependent variable and FDI as the independent variable. As a result of the study, it was stated that there was no cointegration relationship between the variables and, similarly, no causal relationship was found between the variables. While Aslan (2023) and Baydaş & Polat (2018) report no cointegration or causal relationship between FDI and the BIST 100 index, other studies identify different patterns. Aydın & Aksoy (2023) examined the relationship between the BIST 100 and FDI between the periods 2005/4 and 2021/4. According to the ARDL bounds test findings in the study, a long-term relationship was found between the BIST 100 index, FDI, and the dollar exchange rate, and this relationship was statistically significant at the 5% significance level. Furthermore, no causal relationship was found between the variables in the study. Geyikçi (2017), on the other hand, analyzed the impact of FDI on Turkish stock markets between 1986 and 2016 and found long-term cointegration and mutual interaction. As an indirect relationship, Atik (2020) found a bidirectional relationship between the BIST 100 and the foreign share ratio, while Erdem et al. (2025) state that the ratio of foreign investors in the context of the BIST 30, BIST 50, and BIST 100 indices is sensitive to exchange rates, CDS premiums, and stock market performance, but this sensitivity differs according to the indices.

In the international literature, it is also seen that the relationship between FDI and stock market development varies according to country, period, and methodology. Rajapakse (2018), in his study on Sri Lanka, examined both short-term and long-term causal relationships between FDI and the stock market and identified a causal relationship from the stock market to FDI. The study states that policymakers should aim to develop the stock market in order to increase FDI flows to the country. Similarly, in a study covering developing countries, Soumaré & Tchana (2015) state that there is a bidirectional causality between FDI and stock market development, meaning that FDI both develops the stock market and developed financial markets increase FDI. In another study in Vietnam, Vo (2021) states that there is an asymmetric relationship between FDI inflows and the stock market index, and that this relationship can strengthen or weaken depending on changes in capital flows. In another study, Yıldırım & Yıldırım (2025) examined the relationship between stock market development and FDI. Annual data from 2000 to 2020 for Türkiye, Brazil, India, Indonesia, Mexico, and South Africa were analyzed using panel data techniques. The study results show that FDI does not have a significant effect on stock market development. Panel causality tests revealed a unidirectional causality from FDI to stock market development. Zeren & Kılıç (2020), in their 2020 study on G20 countries (excluding Australia and South Korea), found that FDI had no effect on stock markets during the 2013–2019 period. Arčabić et al. (2013) stated that there was no significant long-term relationship between the two variables for Croatia, but that stock market movements could be a determining factor for FDI in the short term.

The literature shows that the relationship between FDI and the stock market varies depending on the country, period, and methodology. While most studies in Türkiye focus on general market indices, short-term relationships between sectoral FDI flows and sector index returns have been examined to a limited extent. Studies have indicated a strong bidirectional relationship between FDI and the stock market in some countries, while in others, weak and

insignificant relationships have been found between the two variables. Similar studies conducted specifically on Türkiye have also found that the direction and strength of the relationship between the two variables can vary depending on the periodic conditions and the methods used. Studies conducted within this scope have generally examined the relationship between the two variables through general market indicators and the BIST 100 index, while sector-level studies are quite limited. However, since the sectoral distribution of FDI differs from the financial and structural characteristics of the sectors, examining this relationship on a sector-by-sector basis will provide more comprehensive and explanatory results. Therefore, this study analyzes the short-term relationships between sectoral FDI and sectoral stock market returns, aiming to fill the gap in the literature on this subject.

Aggregate FDI measures may conceal sectoral heterogeneity because the determinants and location patterns of FDI differ across economic activities. Yu & Walsh (2010) show that the determinants of FDI vary across primary, secondary, and tertiary sectors, while Deichmann et al. (2003) document that FDI location determinants in Türkiye differ substantially across broad industrial categories. Sector-level inflows may therefore convey more targeted information about expected conditions in the corresponding industries than aggregate FDI. From an informationally efficient market perspective, publicly available information relevant to expected firm values should be reflected in stock prices (Fama, 1970). However, limited investor attention affects how information is processed, and investors may devote more attention to market- and sector-wide information than to firm-specific information, with implications for return predictability (Peng & Xiong, 2006). These arguments do not establish that sectoral FDI predicts stock returns. Rather, they motivate an empirical examination of whether sectoral FDI flows contain short-run predictive information that may not be visible in aggregate FDI measures. Accordingly, this study tests the lagged relationships between sectoral FDI flows and the corresponding BIST sector returns.

3. Method

3.1 Data Set and Variables

The study uses monthly data for Türkiye from 2010:01 to 2025:11. Monthly frequency was selected as the primary frequency because sectoral FDI flow data are reported monthly and can therefore be matched consistently with monthly sector index returns. The use of monthly observations also provides a sufficiently detailed time structure for examining short-run and lagged predictive relationships while preserving the number of observations required for sector-specific Vector Autoregression (VAR) estimation. In contrast, aggregation to quarterly frequency would substantially reduce the available observations and degrees of freedom, particularly in multivariate models with lagged variables. Nevertheless, the banking model was also re-estimated using quarterly data as a robustness check to assess whether the main result was driven by the monthly sampling frequency. The dataset consists of sectoral FDI flows, BIST sector index returns, BIST 100 returns, and USD/TL return variables. A portion of the variables used in the study are “sectoral FDI flows”. These variables were obtained from the “Distribution of Foreign Residents’ Direct Investments in Türkiye by Sector–Flow (Million USD)” breakdown in the Balance of Payments data in the Central Bank of the Republic of Türkiye (2025b) Electronic Data Distribution System (TCMB-EVDS). First, total FDI inflows were calculated for each sector during the period 2010:01–2025:11, and the sectors were ranked according to their total investment amounts. The services, industrial, manufacturing, finance and insurance, and wholesale and retail trade sectors, which have the highest FDI inflows, were determined to account for approximately 76–77% of the total sectoral FDI inflows during the period under review. These sectors were included in the analysis because they represent a significant portion of FDI flows. The banking sector was also included in the analysis due to its special importance in the relationship between foreign capital movements and financial markets, and because sectoral FDI data can be directly correlated with the BIST Banking Index. Another variable used in the study is “BIST sector index returns”. These variables represent the monthly logarithmic returns of the indices representing the relevant sectors corresponding to FDI. For this purpose, the historical closing prices of the indices were obtained monthly from Investing.com (n.d.).

In the study, the BIST 100 index return and the USD/TL return were used as additional variables representing general market and exchange rate conditions. The BIST 100 return represents general market movements, while the USD/TL return represents changes in exchange rate conditions. Due to the nature of the VAR model, these variables are also considered endogenously along with other variables. The return series included in the study were calculated as follows:

$$r_t = \ln(P_t/P_{t-1}) \quad (1)$$

3.2 Econometric Method

3.2.1. Vector Autoregression model equation

In this study, VAR modeling was used to examine the dynamic and reciprocal relationships between variables. The VAR approach is a multivariate time series model where all variables are considered endogenous and each

variable is affected by both its own lagged values and the lagged values of other variables (Sims, 1980). In this respect, the VAR model allows for the simultaneous examination of relationships between variables without assuming a causal relationship beforehand. Due to the reciprocal interaction structure observed in financial time series, VAR models are widely used, especially in the analysis of relationships between capital flows and stock markets. In this study, a separate VAR model was estimated for each sector. Each model consists of the relevant sectoral FDI flow, the stock market index return representing that sector, the BIST 100 return, and the USD/TL return. Lag lengths were determined using sector-specific information criteria; the estimated models were subjected to stability and residual autocorrelation tests. Granger causality and impulse response results for sector models that did not meet the required diagnostic tests were excluded from the evaluation. Generalized impulse response functions were used to ensure that dynamic responses are not affected by the ordering of variables. Furthermore, one of the main reasons for choosing this method was that it allows for the existence of potentially bidirectional and lagged relationships between FDI flows and sector index returns, instead of a unidirectional relationship. This structure provides the advantage of being able to analyze variables without the need for a dependent-independent distinction, unlike classical single-equation regression models. Many studies in the literature examining the dynamic relationships between foreign capital movements and financial markets prefer the VAR approach (Arik, 2025; Atik, 2020; Ülkü & İkizlerli, 2012).

In general, a two-variable VAR model with p lags is expressed as follows in this study:

$$Y_t = \alpha_0 + \sum_{i=1}^p \alpha_i Y_{t-i} + \sum_{i=1}^p \beta_i X_{t-i} + \varepsilon_{1t} \quad (2)$$

$$X_t = \gamma_0 + \sum_{i=1}^p \gamma_i X_{t-i} + \sum_{i=1}^p \delta_i Y_{t-i} + \varepsilon_{2t} \quad (3)$$

In this study, a separate VAR model was established for each sector, consisting of the relevant sectoral FDI flow, sector index return, BIST 100 return, and USD/TL return. The general structure of the four-variable model estimated on a sector basis is shown below:

$$R_t = c + \sum_{i=1}^p A_i R_{t-i} + \sum_{i=1}^p B_i F_{t-i} + \sum_{i=1}^p C_i M_{t-i} + u_t \quad (4)$$

where,

R_t : represents the sector index return;

F_t : represents the sectoral FDI flow;

M_t : represents additional variables representing general market and exchange rate conditions (BIST 100 and USD/TL returns);

p : represents the lag length;

u_t : represents the error term.

The lag length used in the model was determined using information criteria, and a lagged structure was preferred in the fundamental analyses. However, alternative lag structures were also tested to examine whether the results were sensitive to the lag length.

3.2.2. Granger causality test

In this study, the Granger Causality Test was applied to examine the bidirectional relationship between variables within the framework of the VAR model (Granger, 1969). The null hypothesis of this test is “The past values of variable X do not explain variable Y.” If the joint significance of the relevant coefficients is statistically rejected, it is assumed that X has Granger causality on Y. This approach was preferred in the study because it provides a suitable framework for evaluating whether FDI flows have predictive power on sector index returns.

3.2.3. Impulse-response analysis

The impulse-response function, another important output of the VAR model, is another analysis preferred in this study. This analysis allows the effect of a one-standard-deviation shock in one variable on other variables to be monitored over time (Sims, 1980). In this study, generalized impulse-response functions were used to prevent the results from being sensitive to the order of variables. The responses of the relevant sector index return to a one-standard-deviation shock in sectoral FDI flows were examined over 12 periods, and the results are presented with 95% confidence intervals.

4. Findings

4.1 Descriptive Statistics

In this section of the study, the descriptive statistical results for the variables used in the analysis are presented in Table 1 and Table 2. In this context, mean, standard deviation, minimum, maximum, skewness, and kurtosis values were calculated for sectoral FDI flows, sector index returns, and general market and exchange rate variables.

According to the descriptive statistics on sectoral FDI flows presented in Table 1, the services sector has the highest average FDI flow, while the wholesale–retail sector has the lowest flow. Standard deviation values are quite high across all sectors, indicating significant fluctuations in FDI inflows over time. Positive skewness and kurtosis values across all sectors show that the distributions exhibit a right-skewed and thick-tailed structure, meaning that exceptionally high investment inflows occurred at specific times. These findings reveal that FDI flows have a heterogeneous and variable structure across sectors.

Table 1. Sectoral foreign direct investment flows–descriptive statistics

Variable	Mean	Standard Deviation	Minimum	Maximum	Skewness	Kurtosis	Observations
Banking	98.811	361.370	0.000	4010.000	8.021	79.462	190
Finance & Insurance	149.789	377.461	0.000	4028.000	7.060	65.359	190
Services	405.905	431.762	57.000	4170.000	4.539	34.251	190
Manufacturing	196.905	242.306	14.000	2038.000	4.304	26.982	190
Industrial	289.937	371.965	19.000	3096.000	4.238	25.879	190
Wholesale–Retail	97.826	163.716	3.000	1437.000	4.376	28.723	190

Table 2 shows descriptive statistics for sector index returns and general market and exchange rate variables. While average returns are similar across sectors, low standard deviations indicate relatively limited fluctuations in index returns. Skewness and kurtosis values are particularly pronounced in the USD return variable, revealing that this variable has a more volatile structure compared to the others.

Table 2. Sector index returns and general market and exchange rate variables–descriptive statistics

Variable	Mean	Standard Deviation	Minimum	Maximum	Skewness	Kurtosis	Observations
Banking index	0.0189	0.1050	-0.2066	0.5179	0.671	4.742	190
Insurance index	0.0254	0.0978	-0.2227	0.5146	1.898	10.506	190
Trade index	0.0226	0.0728	-0.1577	0.2994	0.595	3.622	190
Services index	0.0201	0.0721	-0.1310	0.3015	0.795	4.660	190
Industrial index	0.0211	0.0718	-0.1892	0.2782	0.418	4.094	190
BIST 100	0.0186	0.0748	-0.1543	0.2531	0.501	3.443	190
USD/TRY return	0.0190	0.0526	-0.0786	0.4030	3.525	24.073	190

4.2 Unit Root Test Findings

To ensure the reliability of the VAR model to be applied with the time series variables used in the analysis, the stationarity properties of the series were examined. For this purpose, the Augmented Dickey–Fuller (ADF) unit root test was applied (Dickey & Fuller, 1979). The test results were evaluated separately for the level and first difference values of the variables. The results are presented in Table 3 below:

Table 3. Augmented Dickey–Fuller (ADF) unit root test results

Variable	ADF <i>t</i> -Statistic	<i>p</i> -Value
Foreign Direct Investment Series		
Banking (BANKA)	-13.67	<0.001
Finance and Insurance (FINSGR)	-13.27	<0.001
Services (HIZMTSEK)	-13.27	<0.001
Manufacturing (IMALSEK)	-5.40	<0.001
Industrial (SINAISEK)	-2.78	0.0633
First Difference of Industrial FDI (DSINAI)	-10.44	<0.001
Wholesale and Retail (TOPTPERK)	-8.56	<0.001
Sector Index Returns		
Banking index (XBANKG)	-13.40	<0.001
Insurance index (XSGRTG)	-10.06	<0.001

Trade index (XTCRTG)	-12.07	<0.001
Services index (XUHIZG)	-12.29	<0.001
Industrial index (XUSINAIG)	-11.84	<0.001
General Market and Exchange Rate Variables		
USD/TRY return	-12.39	<0.001
BIST 100 return	-12.41	<0.001

According to the ADF unit root test results presented in Table 3, the p -values of the variables used in the analysis, except for the industrial sector FDI series, are below 0.05 at the level. This result indicates that the series are stationary at the level. The ADF test statistic for the industrial sector FDI series was calculated as -2.78 and the p -value as 0.063. Accordingly, the null hypothesis stating that the series contains a unit root could not be rejected at the 5% significance level. Therefore, the DSINAI variable was created by taking the first difference of the industrial sector FDI series. The ADF test applied to the DSINAI series resulted in a test statistic of -10.44 and a p -value of 0.000. Thus, it was concluded that the industrial sector FDI series is stationary at the first difference. The DSINAI variable, which is stationary in the industrial sector model, was used.

4.3 Lag Length and Model Diagnostic Tests

Information criteria did not always suggest the same lag length for sector models. In the banking model, the Akaike information criterion (AIC), Schwarz information criterion (SC), Hannan–Quinn information criterion (HQ), and final prediction error (FPE) suggested a no-lag structure, while the sequential modified likelihood ratio (LR) test suggested a lagged structure. In the finance and insurance model, the AIC, FPE, and LR criteria supported the VAR(1) model, while the SC and HQ criteria suggested a no-lag structure. Since no-lag models do not allow for the examination of dynamic predictive relationships, the VAR(1) structure, which is the simplest structure among lagged models, was also estimated. Granger causality and impulse response results were interpreted cautiously on the basis that the VAR(1) models satisfy the stability and residual autocorrelation conditions. The results regarding the evaluation of the models are presented in Table 4.

Table 4. Evaluation of sector-specific Vector Autoregression (VAR) models

Sector	Model	Diagnostic Test Result	Use Status
Banking	VAR(1)	Stable, no autocorrelation	Included in the main analysis
Finance and insurance	VAR(1)	Stable, no autocorrelation	Included in the main analysis
Manufacturing	VAR(3)	Stable, with an isolated autocorrelation warning	Excluded from the analysis
Services	VAR(1) and VAR(2)	Autocorrelation persisted	Excluded from the analysis
Wholesale and retail	VAR(1) and VAR(3)	Autocorrelation persisted	Excluded from the analysis
Industrial	VAR(2) and VAR(4)	Autocorrelation persisted	Excluded from the analysis

The banking, finance, and insurance models have met the necessary conditions for stability and residual autocorrelation. Although alternative lag structures were examined in other sector models, residual autocorrelation could not be completely eliminated. Therefore, Granger causality and impulse-response results for these sectors have not been interpreted. This conservative approach was adopted to avoid drawing Granger-causality and impulse-response inferences from VAR specifications that did not satisfy the required stability and residual-diagnostic conditions (Lütkepohl, 2005).

After determining that the series are largely stationary using unit root tests, a VAR model was constructed to examine the dynamic relationships between the variables. The VAR model is a multivariate time series approach where all variables are considered endogenous, and each variable is affected by both its own lagged values and the lagged values of other variables. In this respect, it offers a suitable framework for examining the interaction between foreign investment flows and sector index returns.

4.4 Granger Causality Test Findings

In this study, the Granger Causality Test was applied within the framework of the VAR model to examine the directional relationship between the variables (Granger, 1969). This test reveals whether the past values of one variable statistically explain the current value of another variable. The findings regarding the test results are presented in Table 5 below:

Table 5. Granger causality test results between sectoral foreign direct investment (FDI) flows and index returns

Direction of Causality	<i>p</i> -Value	Result
Banking sector FDI → Banking index return	0.4952	No
Banking index return → Banking sector FDI	0.6768	No
Finance and insurance sector FDI → Insurance index return	0.7734	No
Insurance index return → Finance and insurance sector FDI	0.2471	No

According to the results in Table 5, no statistically significant Granger predictive relationship was found between FDI flows to the banking sector and the banking index return. The reverse relationship was also not significant. Similarly, no significant relationship was found in either direction between FDI flows to the finance and insurance sector and the insurance index return. These findings indicate that sectoral FDI flows do not have a systematic predictive power on the returns of the relevant sector indices in the short term.

In the finance and insurance sector model, a significant Granger relationship was determined from the USD/TL return to the insurance index return ($p = 0.0379$). However, the predictive power of the finance and insurance FDI variable alone on the insurance index return is not statistically significant. This result suggests that exchange rate conditions, rather than sectoral FDI, may be more influential in short-term insurance index movements.

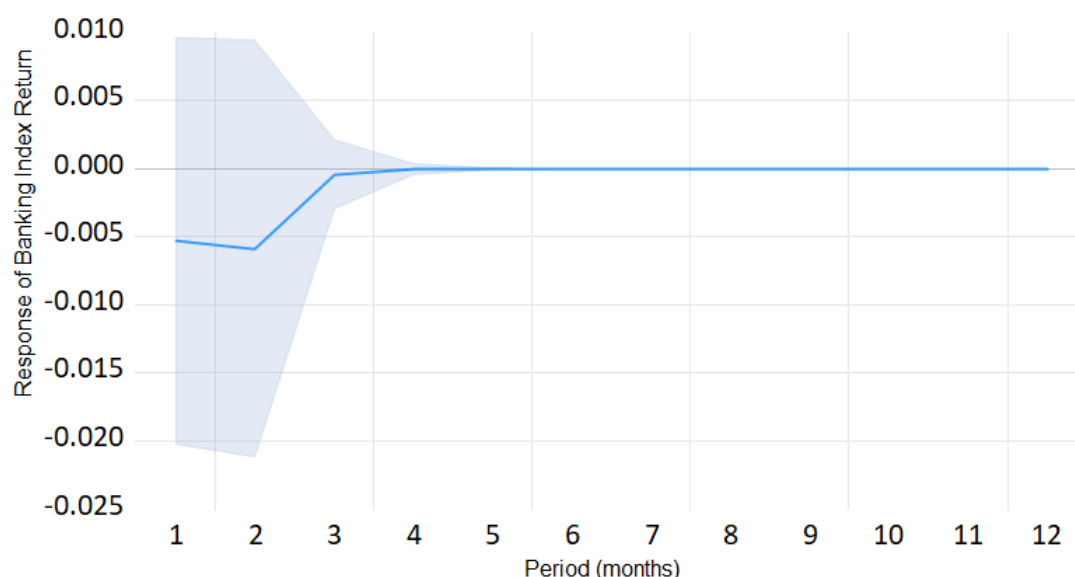
4.4.1. Data frequency reliability check

To examine whether the monthly data results are sensitive to sampling frequency, the banking model was re-estimated using quarterly data. The quarterly model satisfied the conditions for stability and residual autocorrelation. No significant Granger relationship was found between banking sector FDI flows and banking index returns ($p = 0.6965$), and the inverse relationship was also not statistically significant ($p = 0.1272$). Thus, it was observed that the main finding obtained with monthly data was also maintained at the quarterly data frequency.

4.5 Generalized Impulse Response Analysis Findings

Generalized impulse response analysis was applied only to banking and finance and insurance sector models that satisfied the conditions of model stability and residual autocorrelation. The analysis examined the responses of relevant sector index returns to a one-standard-deviation shock in sectoral FDI flows over a 12-month period. Results are presented with 95% confidence intervals.

According to Figure 1, the point estimate of the impact of the FDI shock on the banking sector on the banking index return is only slightly negative in the first two periods. However, the fact that the 95% confidence intervals encompass the zero line in all periods examined indicates that the response is not statistically significant. The convergence of the response to zero from the third and fourth periods onwards suggests that the FDI shock did not have a lasting effect on the banking index return.

**Figure 1.** Response of the banking index return to the banking sector foreign direct investment shock

Note: The figure shows the generalized response of the banking index return to a one-standard-deviation shock in FDI flows to the banking sector. The shaded area represents the 95% confidence interval.

According to Figure 2, the response of the insurance index return to the FDI shock to the finance and insurance sector is only slightly negative around the second period. However, the response is not statistically significant because the confidence intervals encompass the zero line in all periods. The fact that the response weakens and approaches zero over approximately four or five periods indicates that the sectoral FDI shock does not create a lasting price effect on the insurance index return. In general, the index return responses to FDI shocks in the banking and finance and insurance sectors are weak, short-lived, and statistically insignificant. These results are consistent with Granger causality findings showing that sectoral FDI flows do not have a systematic short-term effect on the relevant sector index returns.

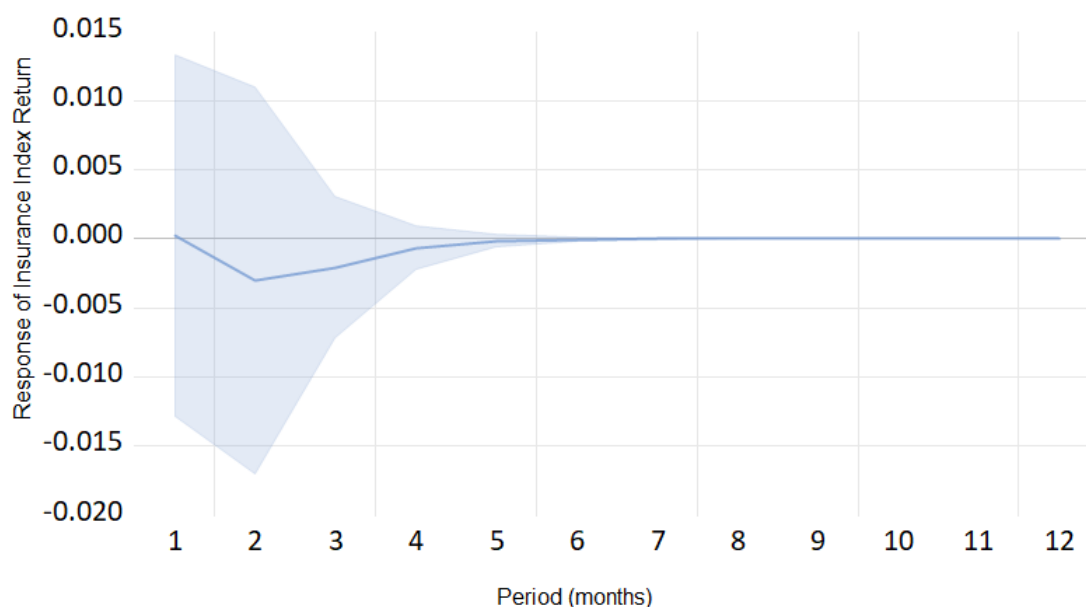


Figure 2. Response of the insurance index return to the foreign direct investment shock in the finance and insurance sectors

Note: The figure shows the generalized response of the insurance index return to a one-standard-deviation shock in FDI flows to the finance and insurance sectors. The shaded area represents the 95% confidence interval.

5. Discussion

The aim of this study is to examine the short-term dynamic relationship between sectoral FDI flows and sector stock market index returns in Türkiye, using monthly data from 2010:01–2025:11. Separate models were developed for the banking, finance and insurance, services, manufacturing, industrial, and wholesale and retail trade sectors. In each model, in addition to the relevant sectoral FDI flow and sector index return, the BIST 100 return and USD/TRY return were used as endogenous variables to represent general market and exchange rate conditions. The estimated models were evaluated in terms of the VAR stability condition and residual autocorrelation; only models meeting the necessary diagnostic conditions were analyzed for Granger causality and generalized impulse response results.

The results of the model diagnostic tests revealed that the banking and finance and insurance sector models met the necessary conditions. However, in the services, wholesale and retail trade, manufacturing, and industrial sector models, despite examining alternative lag structures, the residual autocorrelation problem could not be completely resolved. Therefore, causality and impulse-response findings related to these sectors were excluded from the evaluation. This choice aims to prevent inferences from being made based on estimates that do not meet the model's diagnostic criteria and to increase the reliability of the results.

The exclusion of the services, manufacturing, industrial, and wholesale and retail trade models has important implications for the scope of the findings. The reported Granger-causality and impulse-response results should be interpreted as evidence relating only to the banking and finance and insurance models, rather than as findings representative of all BIST sectors. The failure of the remaining models to satisfy the residual-diagnostic requirements does not demonstrate either the presence or the absence of an FDI–return relationship in those sectors. Instead, it indicates that reliable inference could not be drawn from the current VAR specifications. Accordingly, the conclusions of the study are deliberately limited to the two models that satisfied the required diagnostic conditions and should not be generalized to the excluded sectors.

According to the Granger causality test results, there was no statistically significant predictive relationship from FDI flows to the banking sector to the banking index return. The inverse relationship from the banking index return

to FDI flows to the banking sector is also not significant. Similarly, no significant Granger relationship was found in either direction between FDI flows to the finance and insurance sector and the insurance index return. These findings indicate that in sector models that meet the diagnostic criteria, sectoral FDI flows do not have a strong and systematic short-term predictive power on the returns of the relevant sector index.

The results of the generalized impulse-response analysis also support the Granger causality findings. The point estimate of the banking index return showed a limited negative response in the initial periods to the FDI shock to the banking sector, but the 95% confidence intervals covered the zero line in all periods. In response to the FDI shock to the finance and insurance sector, a short-term and limited negative reaction was observed in the insurance index return, but this reaction was not statistically significant. The fact that the reactions weakened and approached zero over several periods in both models indicates that sectoral FDI shocks do not create a lasting short-term price effect on the returns of the relevant sector indices.

To assess whether the banking sector finding obtained with monthly data is sensitive to data frequency, the model was re-estimated with quarterly data. The quarterly banking model was stable and showed no evidence of residual autocorrelation. No significant Granger relationship was found between banking-sector FDI flows and banking index returns in either direction. Thus, it was observed that the main finding regarding the banking sector is not solely due to monthly data frequency and is maintained in the quarterly analysis.

The findings of this study are consistent with Baydaş & Polat (2018) and Aslan (2023), who report no cointegration or causal relationship between FDI and the BIST 100 index. The results obtained at the sector level in the current research support these findings within a sectoral framework. Similarly, Zeren & Kılıç (2020), in their analysis covering G20 countries, stated that the effect of FDI on stock markets is generally not significant. Yıldırım & Yıldırım (2025), while not finding a significant long-term effect of FDI on stock market development, detected a unidirectional causality from FDI to stock market development. Therefore, the findings of this study support the idea that the effect of FDI on the stock market may be weak, but do not show that the relationship is completely absent. However, there are also studies in the literature that identify stronger or longer-term relationships between FDI and stock market indicators. Aydın & Aksoy (2023) identified a long-term relationship between BIST 100, FDI, and the dollar exchange rate, but did not find a causal relationship between the variables. Geyikçi (2017), on the other hand, found long-term cointegration and mutual interaction between FDI and the stock market. Atik (2020) identified a bidirectional relationship between foreign investor share and the BIST 100 index. While that study focuses on foreign investor share and the general market, the present study examines sectoral FDI flows and sector returns. The findings therefore address complementary dimensions of the foreign capital–stock market relationship.

Atik & Yılmaz (2021) found mostly limited relationships between changes in foreign investor share and sector indices shows that the impact of foreign capital on the stock market can vary depending on the sectors and the capital indicator used. Alper & Kara (2017), while not directly examining FDI, indicate that stock returns can be explained by their own lagged values and macroeconomic variables. This result indirectly supports the assessment that intra-market and macroeconomic dynamics may have a faster impact on short-term stock market movements than sectoral FDI flows. Arčabić et al. (2013), while not finding a significant long-term relationship for Croatia, detected an effect from the stock market to FDI in the short term. Since long-term relationships are not examined in the present study, the long-term findings of the studies cannot be directly compared. In terms of the short term, the relationship determined by Arčabić et al. (2013) from the stock market to FDI differs from the causality finding obtained in this study.

The identification of a significant Granger relationship between USD/TRY returns and insurance index returns in the finance and insurance sector model suggests that short-term sector returns may react more quickly to changes in exchange rates and general financial conditions than to sectoral FDI flows. While the effects of FDI on production capacity, technology transfer, employment, productivity, and firms' future cash flows mostly emerge in the medium and long term, stock market returns can price changes in exchange rates, interest rates, risk perception, and investor expectations in a shorter period. Therefore, the absence of a significant short-term relationship between FDI and sector returns in this study should not be interpreted as meaning that the economic importance of FDI is limited.

The interpretable results for the banking and finance and insurance sectors should not be regarded as proof that these sectors behaved differently from the excluded sectors, because comparable and diagnostically valid evidence could not be obtained for the latter. Nevertheless, the structure of financial intermediation provides a plausible context for the findings. Foreign participation and FDI in the financial sector can influence domestic markets through competition, operating efficiency, regulation, and supervision, which differ from the production-capacity channels generally associated with FDI in non-financial sectors (Claessens et al., 2001; Goldberg, 2007). In the Turkish context, Çelik (2020) finds that interest-rate changes significantly affect the conditional returns of listed insurance companies, whereas exchange-rate risk primarily affects the volatility rather than the mean of their stock returns. Although this evidence is not directly equivalent to the Granger-causality result obtained in the present study, it supports the broader view that Turkish insurance equities are sensitive to macro-financial conditions. Recent developments also underline the importance of distinguishing sectoral FDI from faster-moving forms of

international finance. The Central Bank of the Republic of Türkiye reports that improved external financing conditions and increasing foreign investor interest supported banks' external funding and subordinated-debt issuance in 2024 (Central Bank of the Republic of Türkiye, 2025a). These developments relate primarily to external financing and investor demand rather than direct investment. They therefore provide a possible explanation for why exchange-rate and broader financial conditions may be reflected in financial-sector returns more rapidly than sectoral FDI flows.

6. Conclusion

When the results are evaluated from a policy perspective, it can be said that FDI incentives should not be seen as a tool to increase short-term stock market performance or sector index returns. It would be more appropriate for policymakers to evaluate FDI policies in line with long-term structural goals such as increasing production capacity, supporting technology transfer, creating qualified employment, developing export capacity, and achieving productivity increases, rather than short-term financial market movements. Furthermore, instead of applying FDI policies uniformly to all sectors, they should be designed on a sector-by-sector basis, taking into account the production linkages, technological intensity, level of external dependence, and value-added potential of each sector.

The findings also have practical implications for financial authorities, market regulators, and investors. Financial authorities should avoid interpreting increases in sectoral FDI flows as immediate signals of stronger sector index performance, particularly when assessing short-term financial-market conditions. Instead, sectoral FDI indicators may be evaluated together with exchange-rate movements, risk conditions, market-wide returns, and other high-frequency financial indicators. For market regulators, the results underline the importance of improving the timeliness, sectoral detail, and comparability of foreign-investment disclosures so that market participants can distinguish direct investment from portfolio flows and external borrowing. Investors should likewise avoid using sectoral FDI flows as stand-alone short-term trading signals. The results suggest that sector returns, especially in financial industries, may respond more rapidly to exchange rate and broader market conditions than to changes in direct investment flows. These implications are limited to the banking and finance and insurance models that satisfied the required diagnostic conditions.

The study has some limitations. First, the research is limited to monthly data for Türkiye from 2010 to 2025. Studies considering longer periods, different economic regimes, or structural breaks may yield different results. Second, although FDI flows are examined on a sector basis, investment types such as greenfield investments, mergers, and acquisitions are not separated within themselves. Third, the VAR approach used focuses on examining short-term dynamic relationships, and the results obtained do not allow for direct inferences about long-term structural relationships. Additionally, the failure of some sector models to meet the diagnostic criteria limits the generalization of the study's results to all sectors.

Future studies that disaggregate FDI flows by investment type, utilize micro-firm data, and include structural breaks in the model could provide more detailed results. Furthermore, comparing different developing countries using panel data methods can contribute to determining the extent to which the relationship between FDI and sector stock market returns is affected by country-specific conditions. Using alternative methods such as cointegration, ARDL, or structural VAR to examine long-term relationships can also provide complementary findings to the literature.

Overall, the study revealed that there is no short-term and systematic predictive relationship between sectoral FDI flows and related sector index returns in the banking and finance and insurance sectors, which meet the diagnostic criteria. Generalized impulse response results also showed that responses to FDI shocks were weak, temporary, and statistically insignificant. The consistency of the quarterly robustness analysis for the banking sector with the monthly results indicates that the main finding is somewhat protected against data frequency. In conclusion, it is considered more appropriate to evaluate sectoral FDI flows through their medium- and long-term real and structural effects rather than short-term stock market returns.

Data Availability

The data supporting the findings of this study are publicly available from the Central Bank of the Republic of Türkiye Electronic Data Delivery System (EVDS) and Investing.com. The processed dataset is available from the corresponding author upon reasonable request.

Conflicts of Interest

The author declares no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this manuscript, the author used generative AI-assisted technology solely for language translation, language editing, and improving the clarity of the text. The author reviewed and revised all AI-assisted content and takes full responsibility for the accuracy, originality, and integrity of the manuscript.

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